结构化学Ⅱ A4

本单位制下，
$\lambda\cdot\upsilon = 3\times10^{17}$
$\lambda\cdot E = 1.2\times10^{5}$
$\lambda\cdot\tilde{\upsilon} = 1\times10^{7}$

1 光与分子相互作用基础

$E = h\nu = \hbar\omega,\quad p = \frac{h}{\lambda} = \hbar k\ (k = \frac{2\pi}{\lambda})$

1.1 光子能量

	核	内层e^-	价e^-	振动	转动	e^-自旋	核自旋
	γ-Rays	X-Rays	UV	Vis	IR	Micro	Radio
λ/nm	1×10^{-3}	1×10^{-1}	1×10^{1}	$(4.0-7.8)\times10^{2}$	3×10^{5}		1×10^{9}
ν/Hz	3×10^{20}	3×10^{18}	3×10^{16}	$(7.5-3.84)\times10^{14}$	1×10^{12}		3×10^{8}
E/kJ/mol	1.2×10^{8}	1.2×10^{6}	1.2×10^{4}	$(3.0-1.53)\times10^{2}$	4×10^{-1}		1.2×10^{-4}
$\tilde{\nu}$/cm^{-1}	1×10^{10}	1×10^{8}	1×10^{6}	$(2.5-1.28)\times10^{4}$	3.33×10^{1}		1×10^{-2}

1.2 黑体辐射

谱能量密度：$\rho_\nu = \frac{1}{2}\varepsilon_0 E_\nu^2,\quad \rho = \int\rho_\nu d\nu = \frac{1}{2}\varepsilon_0 E_0^2\ (J\cdot Hz^{-1}\cdot m^{-3})$

谱功率强度/能流密度：$I_\nu = \rho_\nu c,\quad I = \rho c\ (W\cdot Hz^{-1}\cdot m^{-2})$

Planck 方程（热平衡）：$\rho_\nu(\nu) = \left(\frac{8\pi h\nu}{c^3}\right)\left(e^{h\nu/k_BT}-1\right)^{-1}$

1.3.1 光和二能级分子相互作用（半经典）

$E_1, |\varphi_1\rangle$
$E_0, |\varphi_0\rangle$

把外界电场 $E(r,t)$ 作为含时微扰，以非微扰本征态

计算微扰下的波函数 $\psi(t) \Longrightarrow$ 推导不抄了

Bohr 频率：$\omega_{10} = \frac{E_1-E_0}{\hbar}\qquad \mu = -ed$（奇宇称）

Rabi 频率：$\omega_R = \frac{E_0\mu_{10}}{\hbar}$　跃迁偶极矩：$\mu_{01} = \mu_{10} = \langle\varphi_1|\mu|\varphi_0\rangle$

强场下有 $|a_1(t)|^2 = \sin^2\left(\frac{\omega_R t}{2}\right),\quad |a_0(t)|^2 = \cos^2\left(\frac{\omega_R t}{2}\right)$

弱场下有 失谐 $\Delta = \omega_{10}-\omega_{光}$，广义Rabi 频率 $\Omega = \sqrt{\omega_R^2+\Delta^2}$

且有 $|a_1(t)|^2 = \left(\frac{\omega_R}{\Omega}\right)^2\sin^2\left(\frac{\Omega t}{2}\right),\quad |a_0(t)|^2 = 1-\left(\frac{\omega_R}{\Omega}\right)^2\cos^2\left(\frac{\Omega t}{2}\right)$

1.3.2

$\omega_R\to0$，跃迁概率 $P_{1\leftarrow0} = |a_1|^2 = \left(\frac{\omega_R}{\Delta}\right)^2\sin^2\left(\frac{\Delta t}{2}\right) = \frac{\mu_{10}^2E_0^2}{4\hbar^2}\left[\frac{\sin(\Delta t/2)}{\Delta/2}\right]^2$

又：$\lim_{t\to\infty}\left[\frac{\sin(\Delta t/2)}{\Delta/2}\right]^2 = 2\pi t\,\delta(\Delta) \Rightarrow P_{1\leftarrow0} = \frac{2\pi\mu_{10}^2E_0^2}{4\hbar^2}t\,\delta(\omega-\omega_{10})$

∴ 长时单色光下 $W_{1\leftarrow0} = \frac{dP_{1\leftarrow0}}{dt} \propto \mu_{10}^2E_0^2$，与时间 t 无关

有限时多色光下 $W_{1\leftarrow0} = \frac{\pi\mu_{10}^2}{\varepsilon_0\hbar^2}\rho_\nu(\omega_{10})$

1.4.1 多分子系统的吸收与发射（弱场）

$A\ (s^{-1})$
$B\ (m^3J^{-1}s^{-2})$

吸收：$\frac{dN_1}{dt} = B_{1\leftarrow0}\rho_\nu(\nu_{10})N_0$

受激：$\frac{dN_1}{dt} = -B_{1\to0}\rho_\nu(\nu_{10})N_1$

自发：$\frac{dN_1}{dt} = -A_{1\to0}N_1$

$\rho(\nu_{01}) = \frac{A_{1\to0}}{B_{1\leftarrow0}e^{h\nu_{10}/kT}-B_{1\to0}}$

稳态：$\rho(\nu_{01}) = \frac{8\pi h\nu_{01}^3}{c^3}\left(e^{h\nu_{01}/kT}-1\right)^{-1} \Rightarrow B_{1\leftarrow0} = B_{1\to0},\quad A_{1\to0} = \frac{8\pi h\nu_{01}^3}{c^3}B_{1\leftarrow0}$

$\frac{dN_1}{dt} = B_{1\leftarrow0}\rho_\nu(\nu_{10})N_0 \approx B_{1\leftarrow0}\rho_\nu(\nu_{10})N$

$\frac{d(N_1/N)}{dt} = B_{1\leftarrow0}\rho_\nu(\nu_{10})$

各向异性：$W_{1\leftarrow0} = \frac{\pi\mu_{10}^2}{3\varepsilon_0\hbar^2}\rho_\nu(\omega_{10})$

$\rho_\nu(\nu) = \frac{d\rho}{d\omega}\frac{d\omega}{d\nu} = 2\pi\rho_\nu(\omega)$

$\Rightarrow B_{1\leftarrow0} = \frac{\mu_{10}^2}{6\varepsilon_0\hbar^2},\quad A_{1\to0} = \frac{8\pi h\nu_{01}^3}{6\varepsilon_0\hbar^2c^3}\mu_{01}^2$

1.4.2 Lambert-Beer

分子单色光吸收截面(m²)：$\sigma_\nu = \frac{2\pi^2\mu_{10}^2\nu}{3\varepsilon_0hc}g(\nu)$　光谱线型函数(s) $g(\nu)$，$\int_0^\infty g(\nu)d\nu = 1$

$I = I_0e^{-\sigma_\nu NL} = I_0 10^{-\varepsilon_\nu mL}$　（σ_ν: m²，N: m⁻³，L: m；ε_ν: L·mol⁻¹·cm⁻¹，m: mol·L⁻¹，L: cm）　∴ $\varepsilon\sim\sigma_\nu\sim\mu_{10}^2$

1.5 光谱线形函数和展宽

均匀：

- 自然寿命展宽，速率常数 $\gamma = 1/\tau_{sp} = A_{1\to0}$
  半高全宽 $\Delta\nu_{1/2} = \frac{\gamma}{2\pi} = \frac{1}{2\pi\tau_{sp}}$
  [洛伦兹] $g(\nu-\nu_{10}) = \frac{\gamma}{(\gamma/2)^2+4\pi^2(\nu-\nu_{10})^2}$
- 碰撞展宽/压力展宽，$\Delta\nu_{1/2} = bp = \frac{1}{\pi T_2}$
  T_2为碰撞间隔

不均匀：多普勒展宽　$\nu' = \nu(1\pm v/c)$ [平行，同向－，反向＋]

[高斯]　$P_v dv = \left(\frac{m}{2\pi kT}\right)^{1/2}e^{-mv^2/(2kT)}dv$

$g_D(\nu-\nu_0) = \frac{1}{\nu_0}\left(\frac{mc^2}{2\pi kT}\right)^{1/2}e^{-mc^2(\nu-\nu_0)^2/(2kT\nu_0^2)}$

半高全宽：$\Delta\nu_D = 2\nu_0\sqrt{\frac{2kT\ln2}{mc^2}}$

常温常压下，多普勒展宽，压力展宽 >> 自然展宽

2 原子结构和原子光谱

2.1 氢原子电子结构

$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_N}\nabla_N^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}$
$= -\frac{\hbar^2}{2M}\nabla_{C.M.}^2 - \frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}$

$M = m_e+m_N,\quad \mu = \frac{m_em_N}{m_e+m_N} \approx m_e$

波-奥近似：固定原子核，只动电子
$\hookrightarrow \left(-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right)\psi = E\psi$

$E = -\frac{Z^2\mu e^4}{32\pi^2\varepsilon_0^2\hbar^2}\frac{1}{n^2}$，对H：$E = -hcR_H\frac{1}{n^2} = -13.6\frac{1}{n^2}(eV)$

$R_H = \frac{\mu e^4}{8c\varepsilon_0^2h^3} = 109667\ cm^{-1}$

量子数：
- n：主　$n = 0,1,2,\cdots$　简并度 $k = n^2$
- l：角　$l = 0,1,2,\cdots,n-1$
- m_l：磁　$m_l = 0,\pm1,\pm2,\cdots,\pm l$
- m_s：自旋　$m_s = \pm\frac{1}{2}$

$\Psi(n,l,m_l,m_s) = R_{n,l}(r)Y_{l,m_l}(\theta,\varphi)\psi_{m_s}(\sigma)$

$\mu_{if} = \langle\psi_i|\hat{\mu}|\psi_f\rangle,\quad \hat{\mu} = \begin{pmatrix}r\sin\theta\cos\varphi\\ r\sin\theta\sin\varphi\\ r\cos\theta\end{pmatrix}$

$\Rightarrow$ 跃迁选律　$\Delta m_s = 0,\ \Delta l = \pm1,\ \Delta m_l = 0,\pm1$　$\Delta n\in\mathbb{Z}$

2.2 旋轨耦合

$\hat{H}_{SO} = A(r)\vec{l}\cdot\vec{s},\quad A(r) = \frac{1}{4\mu^2c^2}\left(\frac{Ze^2}{4\pi\varepsilon_0r^3}\right)$

$\Rightarrow (\hat{H}_0+\hat{H}_{SO})\psi = E\psi$，一阶微扰法处理

$\hat{j}^2\psi = \hbar^2j(j+1)\psi,\quad j = l+\frac{1}{2},|l-\frac{1}{2}|$

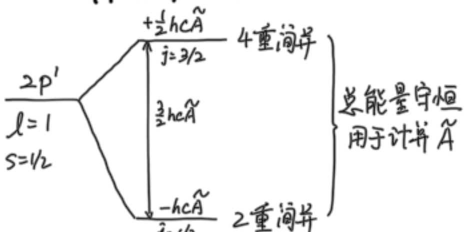
$\hat{j}_z\psi = \hbar m_j\psi,\quad m_j = j, j-1,\cdots,-j$

$\because \vec{j} = \vec{s}+\vec{l}$，$\therefore \hat{H}_{SO} = \frac{1}{2}A(r)(\hat{j}^2-\hat{s}^2-\hat{l}^2)$

$\therefore E_{SO} = \langle lsjm_j|\hat{H}_{SO}|lsjm_j\rangle$
$= \frac{1}{2}hc\tilde{A}\,[j(j+1)-l(l+1)-s(s+1)]$

旋轨耦合常数 (cm⁻¹)　$\tilde{A}\propto Z^4$

例：2p，$l=1$，$s=1/2$



2.3 多电子原子

$\hat{H} = -\frac{\hbar^2}{2m_e}\sum_i^N\nabla_i^2 - \sum_i\frac{Ze^2}{4\pi\varepsilon_0r_i}$

Pauli 原理：交换反对称，$\psi(1,2) = -\psi(2,1)$

Slater 行列式：
$$\psi = (N!)^{-1/2}\begin{vmatrix}\phi_1(1)\alpha(1) & \phi_1(2)\alpha(2) & \cdots & \phi_1(N)\alpha(N)\\ \phi_1(1)\beta(1) & \phi_1(2)\beta(2) & \cdots & \phi_1(N)\beta(N)\\ \phi_2(1)\alpha(1) & \phi_2(2)\alpha(2) & \cdots & \phi_2(N)\alpha(N)\\ \vdots & & & \vdots\\ \phi_{N/2}(1)\beta(1) & \phi_{N/2}(2)\beta(2) & \cdots & \phi_{N/2}(N)\beta(N)\end{vmatrix}_N$$

若N为奇数，末项为(N+1)/2

L-S 耦合（N<57）：La 以前

Step 1：$L = l_1+l_2, l_1+l_2-1,\cdots,|l_1-l_2|$
$S = s_1+s_2, s_1+s_2-1,\cdots,|s_1-s_2|$

Step 2：$J = L+S, L+S-1,\cdots,|L-S|$

J-J 耦合（N>57）

Step 1：$\vec{j_i} = \vec{l_i}+\vec{s_i}$

Step 2：$\vec{J} = \sum_i\vec{j_i} \xleftrightarrow{2e^-} j_1+j_2,\cdots,|j_1-j_2|$

光谱项：$^{2S+1}L_J$

判断光谱项合理性：Pauli 原理

e.g. 同轨道 P^2：$l_1 = l_2 = 1,\ s_1 = s_2 = \frac{1}{2}$
$\Rightarrow L = 2,1,0\quad S = 1,0$

由Pauli：$S=1\leftrightarrow L=1\quad S=0\leftrightarrow L=0,2$

∴ 光谱支项有：$^3P_{2,1,0}\quad ^1D_2\quad ^1S_0$

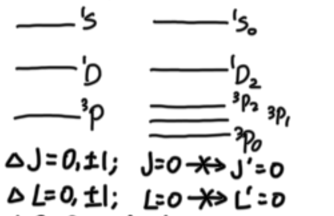
判断光谱支项能量：Hund 规则

Step 1：S↗，E↘

Step 2：同S，L↗，E↓

Step 3：未半满，J↓，E↓
过半满，J↗，E↓

e.g.　p^2　C原子



跃迁选律　$\Delta J = 0,\pm1;\ J=0 \not\leftrightarrow J'=0$
$\Delta L = 0,\pm1;\ L=0 \not\leftrightarrow L'=0$
$\Delta S = 0,\ \Delta l = \pm1$

3 转动光谱

3.1 转动惯量　$I = \sum_i m_ir_i^2$

$[A \overset{R}{-} B \overset{R'}{-} C]$

双原子：$I = \mu R^2,\quad \mu = \frac{m_Am_B}{m}$

线性三原子：$I = m_AR^2+m_CR'^2-\frac{(m_AR-m_CR')^2}{m}$

球形：① CH_4型：$I = \frac{8}{3}m_AR^2$
② SF_6型：$I = 4m_AR^2$

3.2 线性刚性转子转动能级+光谱　$\hat{H}_{rot} = \frac{\hat{l}^2}{2I}$

$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$

$\Rightarrow E_{rot} = \frac{\hbar^2}{2I}J(J+1)\qquad B = \frac{h}{8\pi^2I}(s^{-1})$（测键长）

$\therefore E_{rot} = hBJ(J+1) = hc\tilde{B}J(J+1)$；

以 cm⁻¹为能量单位，$\tilde{E}_{rot} = \tilde{B}J(J+1)$

其中 J为转动量子数，$J = 0,1,2,\cdots$

$m_J = 0,\pm1,\cdots,\pm J$（简并度 $2J+1$）

$\Rightarrow$ 跃迁偶极 $\mu_{if} = \langle i|\vec{\mu}|f\rangle$（有永久偶极才能跃迁）

$\Rightarrow$ 跃迁选律　$\Delta J = \pm1,\ \Delta m_J = 0,\pm1$　同位素，小峰

$\therefore \Delta E = E_{J+1}-E_J = 2hc\tilde{B}(J+1)$，$B/\tilde{B}\propto\frac{1}{I}\nearrow$

$\Rightarrow$ 特征：等间距谱线，$\Delta\tilde{\nu} = 2\tilde{B}$

光谱包络形状：① $|\mu_{J+1\leftarrow J}|^2 = \left(\frac{J+1}{2J+1}\right)\frac{\mu_0^2}{2} \xrightarrow{J\to\infty} \frac{\mu_0^2}{4}$

② $\frac{N_J}{N_0} = \frac{g_J\exp(-E_J/k_BT)}{g_0\exp(-E_0/k_BT)} = (2J+1)\exp(-hBJ(J+1)/k_BT)$

取 $\frac{d(N_J/N_0)}{dJ} = 0$ 有：$J_{max} = \sqrt{\frac{k_BT}{2hc\tilde{B}}}-\frac{1}{2} \Rightarrow$ 测 T

实际：离心畸变 $\Rightarrow I\nearrow,\ \tilde{B}\downarrow,\ \Delta\tilde{\nu}\downarrow$

$\tilde{E} = \tilde{B}J(J+1)-\tilde{D}[J(J+1)]^2,\quad \tilde{D} = 4\tilde{B}^3/\tilde{\nu}^2$，

$\tilde{\nu}$为键振动频率

$\therefore \tilde{\nu}(J+1\leftarrow J) = 2\tilde{B}(J+1)-4\tilde{D}(J+1)^3$，J↗，$\tilde{E}$收敛

3.3 球形转子：同线性转子，但 μ=0，无跃迁

3.4 对称转子：

记 a为分子对称主轴，$\hat{H}_{rot} = \frac{\hat{l}^2}{2I_b}+\hat{L}_a^2\left(\frac{1}{2I_a}-\frac{1}{2I_b}\right)$

$E_{rot} = J(J+1)\frac{\hbar^2}{2I_b}+K^2\hbar^2\left(\frac{1}{2I_a}-\frac{1}{2I_b}\right)$，

$\tilde{E}_{rot} = \tilde{B}J(J+1)+(\tilde{A}-\tilde{B})K^2$.

角量子数 $J = 0,1,2,3\cdots$　$K = 0,\pm1,\cdots,\pm J$（2J+1个取值）

对扁平椭球 (NH_3)，$I_a>I_b=I_c$；

对扁长椭球 (CH_3Cl)，$I_a<I_b=I_c$.

∴ 对扁平，$\tilde{A}<\tilde{B}$；对扁长，$\tilde{A}>\tilde{B}$.

⇒ 对扁平，K↗，$\tilde{E}$↓；对扁长，K↗，$\tilde{E}$↗.

跃迁选律　$\mu_0\neq0,\ \Delta J = \pm1,\ \Delta M_J = 0,\pm1,\ \Delta K = 0$

在不考虑离心畸变的情况下，

$\tilde{\nu}_{J+1,K\leftarrow J,K} = 2\tilde{B}(J+1)$，与K无关，简并度 $(2J+1)^2$

考虑离心畸变，

$\tilde{E}_{JK} = \tilde{B}J(J+1)-\tilde{D}_J[J(J+1)]^2+(\tilde{A}-\tilde{B})K^2-\tilde{D}_KK^4-\tilde{D}_{JK}J(J+1)K^2$

$\therefore \tilde{\nu}_{J+1,K\leftarrow J,K} = 2\tilde{B}(J+1)-4\tilde{D}_J(J+1)^3-2\tilde{D}_{JK}(J+1)K^2$

4 振动光谱

4.1 双原子分子振动能级

$\left(-\frac{\hbar^2}{2\mu}\frac{\partial^2}{\partial x^2}+\frac{1}{2}kx^2\right)\psi = E\psi \Rightarrow E_v = (v+\frac{1}{2})\hbar\omega,\quad \omega = \sqrt{\frac{k}{\mu}}$ (J)

振动量子数 $v = 0,1,2,\cdots$　$\tilde{G}_v = (v+\frac{1}{2})\tilde{\nu}$ (cm⁻¹)

选律　电偶极矩随振动发生改变 即 $\left(\frac{\partial\mu}{\partial x}\right)_0\neq0$；$\Delta v = \pm1$

Morse 势：$V(R) = hc\tilde{D}_e(1-e^{-ax})^2,\quad a = (k/2hc\tilde{D}_e)^{1/2}$

式中 $\tilde{D}_e$ 为平衡解离能，即零点能；

D_0为实际解离能

非谐修正：$E_v = (v+\frac{1}{2})\hbar\omega_e-(v+\frac{1}{2})^2\hbar\omega_e\chi_e$ (J)

$\tilde{G}_v = (v+\frac{1}{2})\tilde{\nu}_e-(v+\frac{1}{2})^2\tilde{\nu}_e\chi_e$ (cm⁻¹)

$\chi_e = \frac{a^2\hbar}{2\mu\omega} = \frac{\tilde{\nu}}{4\tilde{D}_e}$　v↗，$\Delta\tilde{G}_v$↓

$\Rightarrow \tilde{G}_0 = \frac{1}{2}\tilde{\nu}_e-\frac{1}{4}\tilde{\nu}_e\chi_e,\quad \Delta\tilde{G}_v = \frac{1}{2}\tilde{\nu}_e-2\tilde{\nu}_e\chi_e(v+1)$

由于非谐性，波函数变形了，

不严格遵守 $\Delta v = \pm1$，出现倍频峰

同时热带跃迁和基带跃迁谱线不再重合，高温明显

4.2 双原子分子振-转光谱

采用刚性转子-谐振子模型，同时考虑振动+转动.

$\hat{H}=\hat{H}_{vib}(R)+\hat{H}_{rot}(\theta,\varphi) \Rightarrow \psi=\psi_{vib}(R)\psi_{rot}(\theta,\varphi)$

$\Rightarrow E_{vr}=(v+\frac{1}{2})\hbar\omega+hBJ(J+1),\ \tilde{S}_{v,J}=\tilde{G}_v+\tilde{F}_J=(v+\frac{1}{2})\tilde{\nu}+J(J+1)\tilde{B}(cm^{-1})$

跃迁选律 $(\frac{\partial\mu}{\partial x})_0\neq0;\ \Delta v=\pm1;\ \Delta J=0,\pm1,\ \Delta M_J=0,\pm1$

其中ΔJ=0的情况只允许有非满壳层的分子，如NO.

P支(E_{low}): $\tilde\nu_P(J)=\tilde S(v+1,J-1)-\tilde S(v,J)=\tilde\nu-2\tilde BJ,\ \Delta J=-1$

Q支(中心): $\tilde\nu_Q(J)=\tilde S(v+1,J)-\tilde S(v,J)=\tilde\nu,\ \Delta J=0$

R支(E_{high}): $\tilde\nu_R(J)=\tilde S(v+1,J+1)-\tilde S(v,J)=\tilde\nu+2\tilde B(J+1),\ \Delta J=+1$

非谐修正(振动激发态，键长↗，I↗，$\tilde B$↓):

$\tilde\nu_P(J)=\tilde\nu_0-J(\tilde B_0+\tilde B_1)-J^2(\tilde B_0-\tilde B_1)$ [J↗ $\Delta\nu_P$↑]

$\tilde\nu_R(J)=\tilde\nu_0+(J+1)(\tilde B_0+\tilde B_1)-(J+1)^2(\tilde B_0-\tilde B_1)$ [J↗ $\Delta\nu_P$↓]

$\rightarrow\begin{cases}\tilde\nu_R(J)-\tilde\nu_P(J)=4\tilde B_1(J+\frac{1}{2})\\ \tilde\nu_R(J-1)-\tilde\nu_P(J+1)=4\tilde B_0(J+\frac{1}{2})\end{cases}\Rightarrow$ 计算 $\tilde B_0,\tilde B_1$

4.3 多原子分子振动: 自由度 3N-5(线性), 3N-6(非线性)

$\tilde G_{q_i}(v)=(v+\frac{1}{2})\tilde\nu_{q_i},\ \tilde\nu_{q_i}=\frac{1}{2\pi c}(k_{q_i}/m_{q_i})^{1/2}$

[k_q-力常数, m_q-有效质量]

正规振动模式应伴随着偶极矩的变化 ⇒ 有红外活性

谐振近似下，对应模式 $\Delta v_{q_i}=\pm1$ ⟵ 选律

5 分子电子光谱

5.1 双原子分子轨道 [LCAO→MO]

玻-奥近似下，只有H_2^+可以精确求解，其余分子轨道由原子轨道的线性组合得到.(MO) $\Psi=\sum_i c_i\psi_i$ (AO)

LCAO: $\Psi=c_1\psi_1+c_2\psi_2$ $\begin{cases}H_{ii}=\langle\psi_i|\hat H|\psi_i\rangle & \text{库伦积分}\\ H_{ij}=\langle\psi_i|\hat H|\psi_j\rangle & \text{转移}\\ S_{ij}=\langle\psi_i|\psi_j\rangle & \text{重叠}\end{cases}$

$\Rightarrow\begin{vmatrix}H_{11}-E & H_{12}-ES_{12}\\ H_{21}-ES_{21} & H_{22}-E\end{vmatrix}=0\Rightarrow E_\pm=\frac{H_{11}\pm H_{12}}{1\pm S_{12}}\quad \psi_\pm=\frac{\psi_1\pm\psi_2}{\sqrt{2(1\pm S)}}$

$\xrightarrow{\text{近似 } S_{12}\approx0} E_\pm=H_{11}\pm H_{12}\quad \psi_\pm=\frac{\psi_1\mp\psi_2}{\sqrt{2}}$

5.2 双原子分子光谱项

① $^{2S+1}\Lambda^{+/-}_{g/u}$
- Λ: 总轨道角动量量子数(沿z轴)
- S: 总自旋量子数
- g/u: 中心对称性(仅同核双原子)
- +,-: 沿分子轴镜面对称性

$\Lambda=|\sum_i\vec\lambda_i|$, σ→λ=0, π→λ=±1, δ→λ=±2, …

⇒ Λ=0→Σ, Λ=1→Π, Λ=2→Δ, Λ=4→Φ, …

$S=|\sum_i s_i|$. e.g.: H_2^+: S=1/2, H_2: S=0

↳ 在z轴投影，Σ=S, S-1, …, -S+1, -S

若考虑旋轨耦合，Ω=Λ+Σ

② g/u对称性，只考虑同核双原子.

多电子规则: 所有电子MO g/u的乘积

$\boxed{g\times g=g,\ u\times u=g,\ g\times u=u}$

③ +,-对称性: 只考虑未填满电子，总波函数所有的乘积

▲ 所有σ轨道，+对称；π和δ无确定对称性，只需标记Σ光谱项. 不同轨道+,-均可

未成对π电子: 两个π电子三重态为Σ^-，单重态为Σ^+,

e.g.

σ^1	Λ=λ=0	S=s=½	g	⇒ $^2\Sigma_g^+$
π^1	Λ=λ=1	S=s=½	u	⇒ $^2\Pi_u$
$\sigma^1\pi^1$	Λ=1	S=0,1	g×u=u	⇒ $^3\Pi_u$, $^1\Pi_u$
σ^2	Λ=0	S=0	g×g=g	⇒ $^1\Sigma_g^+$
π^4	Λ=0	S=0	u^4=g	⇒ $^1\Sigma_g^+$

闭壳层同核双原子: $^1\Sigma_g^+$; 闭壳层异核双原子分子: $^1\Sigma^+$

Hund规则: S↑ ⇒ Λ↑ ⇒ + < -, E↓

e.g. ① H_2: $(\sigma_g1s)^2$ Λ=0 S=0 $g^2=g$ $(+)^2=(+)$ ⇒ 基态 X $^1\Sigma_g^+$ S_0

$(\sigma_g1s)^1(\sigma_u^*1s)^1$ $\begin{cases}\uparrow\uparrow\ \Lambda=0\ S=1\ g\times u=u\ (+)^2=(+) & b\ ^3\Sigma_u^+\ T_1\\ \uparrow\downarrow\ \Lambda=0\ S=0\ g\times u=u\ (+)^2=(+) & B\ ^1\Sigma_u^+\ S_1\end{cases}$

② O_2 $(\pi_u2p)^4(\pi_g^*2p)^2$: $\lambda_1=\pm1,\lambda_2=\pm1\Rightarrow\Lambda=0,2$; $s_1=s_2=\pm\frac{1}{2}\Rightarrow S=0,1$ ⇒ $^1\Sigma,^3\Sigma,^1\Delta,^3\Delta$

满层，不考虑

又: 在$^3\Delta$中, $\lambda_1=\lambda_2=+1/-1$, $s_1=s_2=+\frac{1}{2}/-\frac{1}{2}$, 在同一轨道中自旋相同，违背Pauli原理，不存在

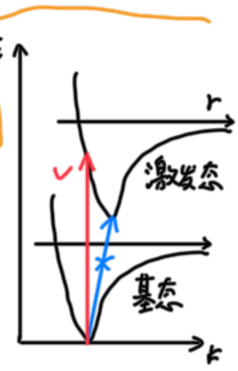
∴ X: $^3\Sigma$, $\lambda_1=+1$, $\lambda_2=-1$, $s_1=s_2=\frac{1}{2}$, $g^2=g$. ⇒ $^3\Sigma_g^-$

a: $^1\Delta$, $\lambda_1=\lambda_2=+1$, $s_1=+\frac{1}{2}$, $s_2=-\frac{1}{2}$. $g^2=g$. ⇒ $^1\Delta_g$

b: $^1\Sigma$, $\lambda_1=+1$, $\lambda_2=-1$, $s_1=+\frac{1}{2}$, $s_2=-\frac{1}{2}$, $g^2=g$ ⇒ $^1\Sigma_g^+$

③ NO: $(\pi^*2p)^1$ Λ=λ=1, S=s=½ ⇒ $^2\Pi$

考虑旋轨耦合: Ω=½/$\frac{3}{2}$ ⇒ $^2\Pi_{1/2}$, $^2\Pi_{3/2}$



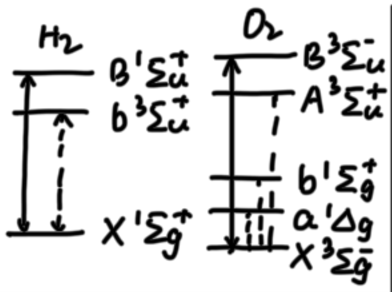
5.3 双原子分子电子跃迁规则

Franck-Condon原理: "垂直跃迁"

跃迁选律 ΔS=0; ΔΛ=0,±1; ΔΩ=0,±1;

由于μ为u对称，g↔u; $\Sigma^+\leftrightarrow\Sigma^+$, $\Sigma^-\leftrightarrow\Sigma^-$

e.g.



5.4 电子-振动光谱

Franck-Condon因子: $S^2_{v,v'}=\langle\psi_v|\psi_{v'}\rangle^2$

由于垂直跃迁，电子跃迁概率取决于初末态重叠程度.

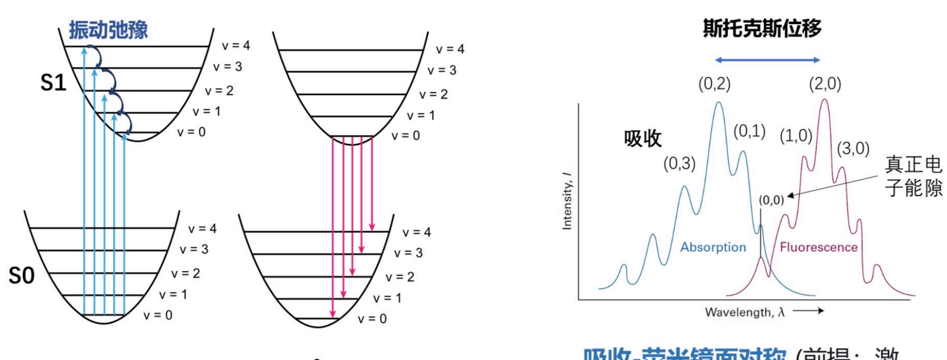
从基态v=0向上跃迁, $S^2_{0,v'}=\langle0|v'\rangle^2=\frac{1}{v'!}(\frac{\Delta^2}{2})^{v'}e^{-\frac{\Delta^2}{2}}$, $\Delta=\sqrt{\frac{\mu\omega}{\hbar}}(R_e'-R_e)$

∴ v=0→v'=0, $S^2_{0,0}=e^{-\frac{\Delta^2}{2}}$

⇒ 峰强影响因素: ①跃迁偶极矩 ②Franck-Condon因子 ③基态Boltzmann分布

Kasha-Vavilov规则: 发光都是由最低激发单重态(S_1)或激发三重态(T_1)的最低振动态发出，与激发波长无关.

吸收-发射光谱



吸收-荧光镜面对称 (前提：激发态和基态振动能级结构相似)

5.5 单-三线态跃迁和磷光

没有旋轨耦合下，S是好量子数，ΔS=0,

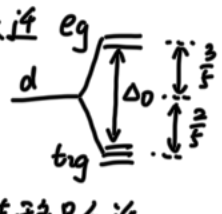
基态为单线态，三线态为光学暗态;

在旋轨耦合下，自旋不是好量子数 ⇝ S-T系间窜跃

T_1成分掺杂在S_1中，跃迁到S_0，发射出磷光.

5.6 多原子分子电子光谱

a) d-d跃迁



①通常在Vis区
②g→g跃迁($e_g\leftrightarrow t_{2g}$)
③较弱(与非对称振动耦合)

b) 电荷转移跃迁

1. 配体→金属(LMCT)
2. 金属→配体(MLCT)

①通常很强，很宽 ②溶剂依赖

c) $\pi^*\leftarrow\pi$和$\pi^*\leftarrow n$跃迁: 取决于对称性；共轭程度↗, λ↗(红移)